BPG-12 · BEST PRACTICE GUIDES

Claims and extension of time

Cause, effect and quantum — and the records that have to exist a year before anyone needs them

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.

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Summary

Delay and disruption are different problems needing different proof, and most submissions fail on causation rather than on quantum. This guide separates the two, sets out the causal chain a claim has to establish, explains what an extension of time does and does not buy, describes the competing approaches to concurrency and to delay analysis without asserting any jurisdiction's position, works a measured-mile disruption calculation, and lists the records that have to exist long before anyone knows a claim is coming. It is not legal advice.

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Contents

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Who this is for. Project controls managers, planners, quantity surveyors and commercial managers on either side of a contract, and project managers who will be asked, eventually, what the records show.

This guide is not legal advice, and it does not state the law of any jurisdiction. Whether a party is entitled to additional time or money, what notice is required, how concurrent delay is treated, and which delay analysis is acceptable are all determined by the specific contract and by the law governing it — and these differ materially between contracts, between countries and between legal traditions. Published industry guidance exists on delay and disruption, of which the Society of Construction Law's Delay and Disruption Protocol is the best known; such guidance is not law, is not binding unless the parties have adopted it, and does not override the contract. Take advice on your own contract from someone qualified to give it.

1. What a claim has to establish

A claim is an assertion of entitlement arising from an event: additional time, additional money, or both. Whatever the contract, the assertion has three parts, and a submission that is strong on one and weak on another fails as a whole.

Cause. An event occurred, and the contract allocates its risk to the other party. This is a contractual question before it is a factual one: an unforeseen ground condition is a cause under some contracts and a contractor risk under others, and the same rainfall is a compensable event, a neutral event or nobody's problem depending on what was agreed.

Effect. The event produced a consequence — days of critical delay, degraded productivity, or both. Most submissions treat this as self-evident, and it is where they are attacked. The effect must be demonstrated by analysis of contemporaneous records, not asserted by proximity in time.

Quantum. The consequence had a cost, evidenced from records, calculated on a stated basis, and not recovered anywhere else.

The order matters. Quantum built before causation is established is a number looking for a reason, and assessors recognise the shape immediately.

2. Delay and disruption are different animals

They are routinely conflated, and they need different proof.

Delay is lateness. The completion date, or a contractual sectional date, moves. The proof is a schedule argument: the event affected the sequence of work that determined the completion date. The remedy is time — an extension — and, separately and conditionally, the time-related cost of the extended period.

Disruption is loss of productivity. The work may finish exactly on time and still cost far more than it should have, because it was performed out of sequence, in congested conditions, with repeated remobilisation, or in short bursts as access was released piecemeal. The proof is a productivity argument, not a schedule one. The remedy is money.

A single event frequently causes both, and the two must then be assessed separately and added carefully — a disrupted, delayed activity can generate a prolongation head and a disruption head, and the risk of counting the same hours twice is severe.

The commonest failure in disruption claims is trying to prove them the way delay is proved. A critical-path analysis says nothing about whether crews were half as productive as they should have been; it only says when things happened. Different question, different evidence.

3. Extension of time: what it buys, and what it does not

An extension of time (EOT) adjusts the contractual completion date. On most contracts it relieves the contractor of liability for liquidated damages (LDs) for the extended period, and it preserves the contractual mechanism, since a completion date that has become impossible is not a date anyone can manage to.

What an EOT does **not** automatically do is pay for the extended period. Time and money are separate questions with separate tests on most contracts, and the outcome depends on the event, the contract and the governing law. It is entirely possible — and on some contracts and some facts, entirely correct — to be granted time and recover nothing.

This asymmetry has a commercial consequence that is often missed by non-commercial readers of a report: an EOT with no money is still valuable, sometimes very valuable, because it removes an exposure that accrues at a fixed rate per day. §9 quantifies both sides of that on the same facts.

Two further points:

Time-related cost is not the same as the LD rate. The LD rate is what the contractor pays for lateness; the prolongation cost is what the contractor actually incurs while remaining on site. They are different numbers with different bases and they should never be substituted for one another.

Acceleration is a third thing. Where an extension is refused or delayed and the contractor accelerates to protect the date, whether the acceleration cost is recoverable — and under what description — varies markedly by contract and jurisdiction, and is contentious in most of them. It should never be assumed, and the decision to accelerate should be documented at the time with its reasons and its instruction status.

4. Concurrency, described without taking sides

Concurrent delay arises when delays for which different parties bear the risk affect the same period. The client's late instruction and the contractor's own resource shortfall each, independently, drive the completion date across the same weeks. The question is what the contractor is entitled to for that period.

Practice recognises several approaches, and different contracts, jurisdictions and forums resolve it differently. Described neutrally:

- **Dominant cause.** The analysis identifies the effective or dominant cause of the delay in the period, which then carries the entitlement. Attractive in principle; frequently impossible in fact, because two genuinely concurrent causes may have no dominant one.
- **Apportionment.** Time, money, or both are divided between the parties in proportion to their causative contribution. Requires a basis for the division that survives challenge.
- **Time but no money.** The contractor receives the extension — relieving the LD exposure — but not the prolongation cost, on the reasoning that the time-related cost would have been incurred anyway because of its own concurrent delay.

There is a fourth question that precedes all of these: **what counts as concurrent at all.** Whether two delays must be truly simultaneous, or merely overlapping in effect, and whether the test is applied to the causes or to their effects, are themselves contested, and different contracts define the term differently or not at all.

Do not assume any of these applies to your project. Some contracts address concurrency expressly; many do not, and then the governing law supplies the answer — and the answer is not the same everywhere. The practical instruction for a controls professional is narrow and useful: **establish the facts, week by week, so that whichever approach applies can be applied to real evidence.** Whichever test a tribunal uses, it will be applied to the records. The party with the better records shapes the analysis, and that advantage is built during the works, not afterwards.

5. Delay analysis at concept level

Several families of method exist for demonstrating the effect of an event on the completion date. Described in principle, and without endorsing any:

- **As-planned versus as-built.** Compare the baseline with what actually happened and identify where the critical sequence diverged. Intuitive and evidence-rich; weak at attributing cause where several things changed at once.
- **Impacted as-planned.** Insert the event into the baseline programme and observe the effect on the computed completion date. Simple; criticised because it tests a plan rather than the project as it was.
- **Time impact analysis.** Insert the event into the programme as it stood immediately before it, updated with actual progress. More faithful to reality; depends entirely on the quality of programme updates.
- **Collapsed as-built.** Remove the delay events from the as-built programme and observe when the works would have completed without them. Powerful where the as-built record is complete; contentious where it is not.

- **Windows or time-slice analysis.** Divide the project into periods and analyse what drove the critical path in each. Usually the most informative where good periodic updates exist, and the most laborious.

Three observations that matter more than the taxonomy:

1. **Different methods can produce different answers on the same facts.** That is a property of the methods, not evidence of manipulation — but it means the choice of method must be justified, on the record, by reference to the contract and the data available.
2. **Every method consumes programme data.** A project with no baseline, or with updates that were never properly statused, cannot run any of them credibly. Schedule quality is therefore a claims asset years before it is a claims issue — see `BPG-05 — Schedule quality: a practical review`.
3. **Some contracts specify the method.** Where they do, that is the end of the discussion.

6. Disruption and the measured mile

Disruption is proved by comparing productivity, and the strongest comparison is against the claimant's own performance on the same work in an un-impacted period — the **measured mile**. The logic is simple and it is why the method persuades: the benchmark is not a published norm or a tender allowance that the other party can attack as optimistic, but what these crews actually achieved on this project.

```
Mile productivity     = hours ÷ quantity, in the un-impacted period
Impacted productivity = hours ÷ quantity, in the impacted period
Excess hours          = (impacted productivity − mile productivity) × impacted quantity
Quantum               = excess hours × demonstrated cost rate
```

The method stands or falls on the comparability of the mile. Four tests before using a period as the mile:

- **Same work** — same activity, specification and materials.
- **Same resources** — same or equivalent crews, same supervision, same shift pattern.
- **Same conditions, apart from the impact** — not the winter period against the summer one.
- **Not the learning curve.** A mile taken from the first work performed is penalised by inexperience, so it understates productivity and overstates the claim — precisely the weakness an assessor looks for. Choose a mature period.

Where no clean mile exists — and often none does — the fallback is a bottom-up build of the lost hours from contemporaneous records, event by event. What must not happen is a leap to a global figure (see §8).

7. Records that survive scrutiny

By the time anyone needs them, the records either exist or they do not. There is no retrospective fix, and the difference between a project with records and one without is usually the difference between a settlement and a write-off.

Programme records. The accepted baseline, every periodic update with its data date, and the statusing that produced it — actual dates, remaining durations, logic changes with reasons. An update re-logicked without a note is worth much less than one that explains itself.

Labour and productivity records. Timesheets or allocation sheets coded to area, activity and — the crucial one — to a change or event reference where applicable, with quantities installed per period in the same coding. Coding at source is what makes a measured mile or a bottom-up build possible later.

Daily records. Site diaries recording resources, work fronts available, weather, access restrictions, instructions received and interruptions. Contemporaneous, dated, and preferably signed.

Correspondence and notices. A notices register showing what was notified, when, under what provision and with what response; instructions with their references and issuers.

Photographs, survey records and meeting minutes — dated and located, showing conditions rather than achievements, and recording access dates and undertakings given.

Two rules govern all of it. **The record is made for the project, not for the claim** — records manufactured once a dispute is in sight are visibly different and are treated as such. And **contemporaneous means contemporaneous**: a diary written up weekly from memory is a reconstruction.

8. How this goes wrong

The global claim. The total cost overrun is presented as the composite consequence of many events, without linking each cause to its own effect and its own quantum. It is tempting where events genuinely interact, and it is fragile: because the quantum is one undifferentiated sum, demonstrating that any part of it was caused by the claimant, or that the tender was underpriced, undermines the whole. There is no mechanism inside a global claim for removing the bad element. The antidote is structural and early — cost coding fine enough to isolate each event's cost as it is incurred, which is a controls decision made a year before anyone contemplates a claim.

Notice missed. Covered in `BPG-11 — Change orders and variations` §2. The consequence of late notice varies by contract and by governing law, and on some contracts it is fatal. It is nearly always an administrative failure rather than a commercial one.

Causation assumed from proximity. "The instruction was issued in Week 14 and we finished eight weeks late" is not an argument. The analysis must show that the event affected the sequence that determined the completion date, in the period it occurred.

Prolongation priced at the LD rate. Convenient, and wrong. Prolongation is what was actually incurred, evidenced; the LD rate is a different number agreed for a different purpose.

Double recovery between variations and claims. The most damaging quantum defect. Hours recovered inside an instructed variation and again in a disruption claim; preliminaries recovered in a star rate and again as prolongation. One instance found invites the assessor to distrust the entire submission, including the parts that were correct.

A measured mile chosen because it produces the best number. An early, short or unrepresentative mile will not survive comparison with the rest of the record. Choose the mile on comparability grounds and record why.

Head office overhead asserted by formula. Formula-based approaches to unabsorbed head-office overhead exist and are used, and their acceptability varies substantially between jurisdictions and forums. Presenting one as though it were settled entitlement invites a challenge that can discredit better-founded heads alongside it.

Records assembled retrospectively. The claim is written first and the evidence is searched for afterwards. It shows: gaps in the coding, diaries with uniform handwriting, and quantities that do not reconcile to the valuations submitted at the time.

Nobody owns the file. Claims are lost by projects where three people each held part of the record and none of them held the chronology.

9. Worked example

Illustrative figures. Currency USD; a single work front on a construction contract; the example assumes that entitlement under the contract has been established for the events described and quantifies the consequence only. **The arithmetic below does not create entitlement; whether these events give any entitlement at all is a question about the contract and the governing law.**

9.1 Prolongation for a 24-day critical delay

A window analysis of the period shows that late release of a work front delayed the sequence determining the completion date by **24 calendar days**. Time-related costs actually incurred and evidenced during the extended period:

Time-related cost head	Rate per calendar day
Site establishment and supervision	9,400
Tower crane hire and operation	2,150
Site accommodation and services	780
Total	**12,330**

```
Prolongation quantum = 12,330 × 24 = 295,920
```

Every element is a cost the project actually incurred because it remained on site, evidenced from invoices, hire records and payroll — not an allowance from the tender, and not the liquidated damages rate.

9.2 The value of the extension itself

Liquidated damages under this illustrative contract run at **18,000 per calendar day**.

```
LD exposure removed by a 24-day extension = 18,000 × 24 = 432,000
```

Set the two side by side, because they answer different questions:

Outcome	Prolongation cost recovered	LD exposure removed	Net position
Extension granted, with cost	295,920	432,000	+727,920
Extension granted, no cost	0	432,000	+432,000
Extension refused	0	0	(432,000) exposure retained

The middle row is the one that surprises people. On facts where the contract gives time but not money — a neutral event, or a concurrency outcome of that kind — the extension is still worth 432,000, because it switches off an exposure priced per day. A report that presents "no money recovered" as a failure has misread the commercial position.

9.3 Disruption: a measured mile

Separately, pipe support installation on the same contract was disrupted by piecemeal release of work fronts. Two periods, same crews, same support type, same specification:

Period	Supports installed	Labour hours	Productivity
Un-impacted (the mile)	2,100	6,720	`6,720 ÷ 2,100 = 3.20 h/support`
Impacted	1,450	6,090	`6,090 ÷ 1,450 = 4.20 h/support`

```
Excess productivity = 4.20 − 3.20 = 1.00 h/support
Excess hours        = 1.00 × 1,450 = 1,450 hours
Demonstrated cost rate (from payroll, all-in) = 62.00 / hour
Gross quantum       = 1,450 × 62.00 = 89,900
```

Now remove what has already been recovered. Of those 1,450 excess hours, **180** relate to work that was also the subject of an instructed variation, priced on a built-up rate that included the additional labour. Recovering them again here would be double recovery:

```
Net excess hours = 1,450 − 180 = 1,270
Net quantum      = 1,270 × 62.00 = 78,740
```

This deduction is the single most valuable paragraph in a disruption submission. Making it yourself, and showing the workings, tells an assessor that the rest of the numbers have been through the same test.

9.4 The total, and what it rests on

```
Prolongation (§9.1)        = 295,920
Disruption, net (§9.3)     =  78,740
Total quantified           = 374,660
```

Assumptions this example depends on. Entitlement under the contract is assumed, not demonstrated; the 24-day critical delay is the output of a window analysis of contemporaneous programme updates, not an assertion; the mile period is comparable on work type, crews, shift pattern and conditions, and is not the first work performed; the cost rate is demonstrated from payroll rather than tendered; no head-office overhead is claimed; and no element of the disruption quantum overlaps with the prolongation head, which covers time-related site costs only.

10. Checklist

Before anything happens — the standing position

- [ ] The baseline programme is accepted and issued, and updates are produced on a fixed cycle with data dates and statusing recorded.
- [ ] Timesheets are coded to area and activity, and can be coded to an event reference on the day one arises.
- [ ] Site diaries are written daily, recording resources, work fronts, access and interruptions.
- [ ] A notices register exists with one owner.
- [ ] Someone is named as owner of the records chronology.

When an event occurs

- [ ] The event is logged with its date on the day it is identified.
- [ ] The contract has been read for the applicable notice requirement, and notice given or diarised.
- [ ] Separate cost codes are opened before the affected work proceeds.
- [ ] Photographs, minutes and correspondence relating to the event are collected into one file immediately.
- [ ] The potential effect — delay, disruption, or both — is stated at the outset, because they need different evidence.

Building the case

- [ ] Cause is established against the contract, not merely as a matter of fact.
- [ ] Effect is demonstrated by an analysis whose method is stated and justified.
- [ ] Where concurrency may arise, the week-by-week factual position is documented before any legal characterisation is attempted.
- [ ] The delay analysis and the disruption analysis are kept separate.
- [ ] Any measured mile is justified against the four comparability tests.

Quantum

- [ ] Prolongation is priced from costs actually incurred, not from the LD rate or tender allowances.
- [ ] Every head is checked against variations already valued, and overlaps deducted with the workings shown.
- [ ] Acceleration, if claimed, has a contemporaneous record of the decision and its reasons.
- [ ] Nothing is claimed that the records cannot support, at any value.

Before submission

- [ ] Each event has its own cause, effect and quantum — nothing is aggregated for convenience.
- [ ] The submission states what it assumes and what it does not.
- [ ] Legal advice has been taken on entitlement and on the governing law.

Related

- `BPG-11 — Change orders and variations` — the valuation route that resolves most changes before they become claims, and the double-counting risk between the two.
- `BPG-05 — Schedule quality: a practical review` — the programme discipline every delay analysis consumes.
- `BPG-04 — Baselining and baseline change control` — why an accepted, controlled baseline is the starting point of any delay argument.
- `TPL-13 — Claim and extension-of-time narrative structure` — the structure for setting out cause, effect and quantum.
- `TPL-12 — Change order log` — the register that becomes the first draft of a claim file.

Sources and standards

Contracts and standard forms — including those published by FIDIC and the NEC family — allocate risk, require notice and define valuation differently from one another, and their operation also depends on the governing law; no clause, wording or numbering from any of them is reproduced here and no jurisdiction's position is asserted. Published industry guidance on delay and disruption exists, of which the Society of Construction Law's Delay and Disruption Protocol is the best known; it is guidance rather than law, is not binding unless adopted by the parties, and its substantive positions are not stated here. Internal references are BoK Domain 7 (Contracts, Commercial Management, BoQ, Invoicing & Revenue) and BoK Domain 10 (Project Scheduling). All figures in §9 are illustrative and were computed for this document.

Status and version

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